USN | | | | | | | | | | [23CSE505]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

ADVANCED DBMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5 x2=10**
1.	**a.**	What is an Object-Relational Database (ORD)?	L1	CO1	**02**
	b.	What is a secondary storage device?	L1	CO2	**02**
	c.	What is concurrency control in the context of distributed databases?	L1	CO3	**02**
	d.	What are NoSQL databases?	L1	CO4	**02**
	e.	What is information retrieval (IR)?	L1	CO5	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Explain how an Object-Relational Database differs from a traditional Relational Database.	L2	CO1	**05**
		OR			
	b.	How would you define a structured data type in an SQL database to represent a customer's address, including fields like street, city, and postal code?	L3	CO1	**05**
3.	**a.**	Describe how data is stored and retrieved from a hard disk drive (HDD).	L2	CO2	**05**
		OR			
	b.	Explain how buffering improves the efficiency of block data transfers between memory and secondary storage.	L2	CO2	**05**
4.	**a.**	Explain the purpose of data fragmentation in distributed databases.	L2	CO3	**05**
		OR			
	b.	Demonstrate how to perform a recovery operation after a system crash in a distributed database scenario.	L3	CO3	**05**
5.	**a.**	Explain the significance of the CAP theorem in the context of distributed systems.	L2	CO4	**05**
		OR			
	b.	Explain the basic structure of a key-value pair in a key-value store.	L2	CO4	**05**
6.	**a.**	Explain in detail about various Retrieval models.	L2	CO5	**05**
		OR			
	b.	List at least three technologies or methodologies that are shaping	L1	CO5	**05**

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		the future of information retrieval.			
7.	**a.**	Discuss in detail about Architecture of Distributed Database.	L2	CO3	**10**
		OR			
	b.	Explain the basic components of a graph database, including nodes, edges, and properties.	L2	CO4	**10**
		PART - C		**6 X 10=60**	
8.	**a.**	Describe the role of complex data types in designing a database for ORDBMS.	L2	CO1	**10**
		OR			
	b.	Describe the relationship between classes and objects in an OODBMS.	L2	CO1	**10**
9.	**a.**	How would you apply the concept of file locking to prevent concurrent access issues in a multi-user environment?	L3	CO2	**10**
		OR			
	b.	Demonstrate how you would implement an insert operation for an unordered record file in a database system.	L3	CO2	**10**
10.	**a.**	Demonstrate the steps involved in coordinating a distributed transaction using a two-phase commit protocol.	L3	CO3	**10**
		OR			
	b.	Explain the Main Phases involved in Query Processing.	L3	CO3	**10**
11.	**a.**	How would you implement a document-based database for a content management system (CMS) that requires flexible schemas?	L4	CO4	**10**
		OR			
	b.	Explain the document model used by MongoDB and how it differs from the traditional relational model.	L4	CO4	**10**
12.	**a.**	Discuss the Concept of Different types of Queries used in IR system.	L2	CO5	**10**
		OR			
	b.	Explain the role of crawlers in web search and how they gather data from the internet.	L2	CO5	**10**
13.	**a.**	Explain the concept of data distribution in distributed databases.	L2	CO3	**10**
		OR			
	b.	Explain the concept of tokenization in text processing.	L2	CO5	**10**